



Eating Disorders

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Introduction

Eating disorders include a variety of afflictions characterized by disturbed eating-related behaviours causing altered consumption of food and impairing health or function.^[1] Among these disorders, anorexia nervosa, bulimia nervosa, binge-eating disorder and avoidant/restrictive food intake disorder are addressed in this chapter.

Anorexia nervosa is characterized by a distorted body image and excessive dieting that leads to severe weight loss and a pathological fear of becoming overweight. Although most patients with anorexia nervosa present initially as restricting type, many also exhibit binge/purge behaviours later in the course of the disease. Many patients will alternate between periods of restricting and bingeing/purging. Anorexia nervosa primarily affects adolescent and young females, but males can also be affected. *(Note: Studies presenting this finding classify patients as female or male. CPhA is aware these terms may not reflect the identity of the individual patient presenting and are not inclusive of all patients. Each patient should be considered individually.)*

Bulimia nervosa is characterized by frequent episodes of binge eating followed by inappropriate behaviours, such as self-induced vomiting, to avoid weight gain. It typically presents in individuals within a normal or overweight range.

Binge-eating disorder is characterized by bingeing episodes during which individuals ingest large amounts of food and experience loss of control over their eating behaviour.

Patients with avoidant/restrictive food intake disorder (ARFID) have poor nutrition and may restrict the quantity or variety of their food intake. The restriction is not motivated by body image or weight concerns, but rather by learned aversion. Aversion may be caused by specific sensory dislikes or negative emotions that previously occurred around the time of eating and may include fear of vomiting, choking or allergic reaction.

See [Table 1](#) for diagnostic criteria from the Diagnostic and Statistical Manual of Mental Disorders, 5th edition, Text Revision (DSM-5-TR).^[1]

Table 1: Diagnostic Criteria for Select Eating Disorders^[1]

Eating Disorder	Diagnostic Criteria
Anorexia Nervosa <i>Specify</i> current severity: The minimum level of severity is based, for adults, on current body mass index (BMI) (see below) or, for children and adolescents, on BMI percentile. The ranges below are derived from World Health Organization categories for thinness in adults; for children and adolescents, corresponding BMI percentiles should be used. The level of severity may be increased to reflect clinical symptoms, the degree of functional disability, and the need for supervision. • Mild: BMI $\geq$ 17 kg/m² • Moderate: BMI 16–16.99 kg/m² • Severe: BMI 15–15.99 kg/m² • Extreme: BMI < 15 kg/m²	A. Restriction of energy intake relative to requirements, leading to a significantly low body weight in the context of age, sex, developmental trajectory, and physical health. <i>Significantly low weight</i> is defined as a weight that is less than minimally normal or, for children and adolescents, less than that minimally expected. B. Intense fear of gaining weight or of becoming fat, or persistent behavior that interferes with weight gain, even though at a significantly low weight. C. Disturbance in the way in which one's body weight or shape is experienced, undue influence of body weight or shape on self-evaluation or persistent lack of recognition of the seriousness of the current low body weight. Coding note: The ICD-10-CM code depends on the subtype (see below). <i>Specify</i> whether: • F50.01 Restricting type: During the last 3 months, the individual has not engaged in recurrent episodes of binge eating or purging behavior (i.e., self-induced vomiting or the misuse of laxatives, diuretics or enemas). This subtype describes presentations in which weight loss is accomplished primarily through dieting, fasting, and/or excessive exercise. • F50.02 Binge-eating/purging type: During the last 3 months, the individual has engaged in recurrent episodes of binge eating or purging behavior (i.e., self-induced vomiting or the misuse of laxatives, diuretics or enemas). <i>Specify</i> if: • In partial remission: After full criteria for anorexia nervosa were previously met, Criteria A (low body weight) has not been met for a sustained period, but either Criterion B (intense fear of gaining weight or becoming fat or behavior that interferes with weight gain) or Criterion C (disturbances in self-perception of weight and shape) is still met.

Eating Disorder	Diagnostic Criteria
	• In full remission: After full criteria for anorexia nervosa were previously met, none of the criteria have been met for a sustained period of time.
Bulimia Nervosa <i>Specify</i> current severity: The minimum level of severity is based on the frequency of inappropriate compensatory behaviors (see below). The level of severity may be increased to reflect other symptoms and the degree of functional disability. • Mild: An average of 1–3 episodes of inappropriate compensatory behaviors per week • Moderate: An average of 4–7 episodes of inappropriate compensatory behaviors per week • Severe: An average of 8–13 episodes of inappropriate compensatory behaviors per week • Extreme: An average of 14 or more episodes of inappropriate compensatory behaviors per week	A. Recurrent episodes of binge eating. An episode of binge eating is characterized by both of the following: 1. Eating, in a discrete period of time (e.g., within any 2–hour period), an amount of food that is definitely larger than what most individuals would eat in a similar period of time under similar circumstances. 2. A sense of lack of control over eating during the episode (e.g., a feeling that one cannot stop eating or control what or how much one is eating). B. Recurrent inappropriate compensatory behaviors in order to prevent weight gain, such as self-induced vomiting; misuse of laxatives, diuretics, or other medications; fasting; or excessive exercise. C. The binge eating and inappropriate compensatory behaviors both occur, on average, at least once a week for 3 months. D. Self-evaluation is unduly influenced by body shape and weight. E. The disturbance does not occur exclusively during episodes of anorexia nervosa. <i>Specify</i> if: • In partial remission: After full criteria for bulimia nervosa were previously met, some, but not all, of the criteria have been met for a sustained period of time. • In full remission: After full criteria for bulimia nervosa were previously met, none of the criteria have been met for a sustained period of time.
Binge-Eating Disorder <i>Specify</i> current severity: The minimum level of severity is based on the frequency of episodes of binge eating (see below). The level of severity may be increased to	A. Recurrent episodes of binge eating. An episode of binge eating is characterized by both of the following: 1. Eating, in a discrete period of time (e.g., within any 2–hour period), an amount of food that is definitely larger than what most individuals would eat in a similar period of time under similar circumstances.

Eating Disorder	Diagnostic Criteria
reflect other symptoms and the degree of functional disability. • Mild: 1–3 binge-eating episodes per week • Moderate: 4–7 binge-eating episodes per week • Severe: 8–13 binge-eating episodes per week • Extreme: 14 or more binge-eating episodes per week	2. A sense of lack of control over eating during the episode (e.g., a feeling that one cannot stop eating or control what or how much one is eating). B. The binge eating episodes are associated with three (or more) of the following: 1. Eating much more rapidly than normal. 2. Eating until feeling uncomfortably full. 3. Eating large amounts of food when not feeling physically hungry. 4. Eating alone because of feeling embarrassed by how much one is eating. 5. Feeling disgusted with oneself, depressed, or very guilty afterward. C. Marked distress regarding binge eating is present. D. The binge eating occurs, on average, at least once a week for 3 months. E. The binge eating is not associated with the recurrent use of inappropriate compensatory behavior as in bulimia nervosa and does not occur exclusively during the course of bulimia nervosa or anorexia nervosa. <i>Specify</i> if: • In partial remission: After full criteria for binge-eating disorder were previously met, binge-eating occurs at an average frequency of less than one episode per week for a sustained period of time. • In full remission: After full criteria for binge-eating disorder were previously met, none of the criteria have been met for a sustained period of time.
Avoidant/Restrictive Food Intake Disorder	A. An eating or feeding disturbance (e.g., apparent lack of interest in eating or food; avoidance based on the sensory characteristics of food; concern about aversive consequences of eating) associated with one (or more) of the following: 1. Significant weight loss (or failure to achieve expected weight gain or faltering growth in children). 2. Significant nutritional deficiency. 3. Dependence on enteral feeding or oral nutritional supplements.

Eating Disorder	Diagnostic Criteria
	4. Marked interference with psychosocial functioning. B. The disturbance is not better explained by lack of available food or by an associated culturally sanctioned practice. C. The eating disturbance does not occur exclusively during the course of anorexia nervosa or bulimia nervosa, and there is no evidence of a disturbance in the way in which one's body weight or shape is experienced. D. The eating disturbance is not attributable to a concurrent medical condition or not better explained by another mental disorder. When the eating disturbance occurs in the context of another condition or disorder, the severity of the eating disturbance exceeds that routinely associated with the condition or disorder and warrants additional clinical attention. <i>Specify if:</i> • In remission: After full criteria for avoidant/restrictive food intake disorder were previously met, the criteria have not been met for a sustained period of time.
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Goals of Therapy

- Assess and treat coexisting nutritional deficiencies
- Assess and treat the effects of malnutrition, e.g., osteoporosis, hypoglycemia, dehydration, electrolyte imbalance
- Improve cognitive and emotional function
- Identify and treat family dysfunction and/or psychiatric comorbidities, e.g., obsessive compulsive rumination, anxiety, depression, self-injurious behaviour, suicidal ideation
- Reduce or eliminate patient's binge and purge behaviour
- Encourage healthy eating habits
- For anorexia nervosa and ARFID, reach and maintain a healthy weight
- For ARFID, desensitize to foods (textures, tastes or smells) while ensuring the goals of a healthy diet and allowing the patient to eat socially with friends

- Prevent relapse

Investigations

- A thorough history with special attention to:
 - weight; eating habits; binge and purge behaviour; menstruation (if appropriate); body image; use of vomiting, laxatives (oral formulations or rectal suppositories), diuretics, ipecac, products marketed as weight loss supplements; fasting; excessive exercising
 - depression, anxiety, suicidal ideation, family dysfunction, sexual abuse
 - symptoms of malnutrition, including chest wall pain, palpitations, loss of consciousness, hematemesis, seizures, abdominal pain, muscle weakness or cramping, episodes of confusion
 - symptoms of autonomic dysfunction (which increase the chance of arrhythmia): episodic sweating, change in heart rate, blood pressure or dizziness on standing, palpitations, sudden need to defecate or urinate
 - developmental and psychological history
 - dietary history, including the patient's dietary goals (e.g., foods they would like to eat to fit in with their friends) and any foods, textures, tastes or smells they find aversive
- Physical examination for parotid hypertrophy, jugular venous pressure, edema, abnormal dentition, Russell sign (scarring over the knuckles caused by using the hand to induce vomiting); for anorexia nervosa and ARFID, examine for postural hypotension, tachycardia, lanugo hair, hypercarotenemia, change in height and/or weight, measurements of body fat, proximal myopathy, neuromuscular hyperirritability (Chvostek and Trousseau signs)^[2]
- Laboratory tests: see [Table 2](#). Note that electrolyte abnormalities in patients with eating disorders are associated with a higher risk of hospitalization, poor health outcomes and mortality^[3]
- For anorexia nervosa and ARFID, a psychiatric and nutritional assessment if symptoms continue and weight does not normalize with 1–2 months of weekly follow-up and counselling^[4]

Note that body weight corrected to height (BMI) is often used as a measure of adiposity but has several limitations. This classification system was created using data from White European males and is less accurate with respect to young adults who have not yet reached full growth, lean or muscular adults, adults over 65 years of age, and people of different races.^{[5][6][7]}

Table 2: Recommended Laboratory Tests for Eating Disorders

Laboratory Test	Related to Nutritional Restriction	Related to Purging
Sodium, potassium, chloride, bicarbonate, magnesium, calcium, phosphorus and zinc	x	x
Creatinine	x	x
Vitamin B ₁₂	x	x
Ferritin	x	x
ECG	x	
Hemoglobin, RBC and WBC count	x	x
Urinalysis	x	
Folate	x	
Blood glucose	x	
Liver function tests	x	

Abbreviations: ECG = electrocardiogram; RBC = red blood cell; WBC = white blood cell

Anorexia Nervosa

Therapeutic Choices

[Figure 1](#) illustrates the management of anorexia nervosa. Body fat must be normalized for psychological treatment to be effective and for restoration of normal psychological and physical function.^[8] Since normal body fat may differ among body types, other appropriate measures include body weight and/or

percentile on an appropriate growth chart. Realistic targets for weight gain are 0.9–1.8 kg per week in hospitalized patients or those in residential programs, 0.4–1.4 kg per week for partially hospitalized patients, and 0.4–0.9 kg per week for patients in outpatient programs, until a normal body fat percentage, body weight or growth chart percentile is achieved.^[8] Additional therapeutic targets include:

- Continued linear growth in those whose epiphyses have not closed
- Menstruation (although this may start or resume due to stress)
- Ovulation (as demonstrated by dominant ovarian follicles on ultrasound)
- Decreased fear of weight gain
- Normalization of strength
- Normalization of cognitive function
- BMI

Nonpharmacologic Choices

- Develop and maintain a rapport and therapeutic alliance.
- Recommend psychotherapy, including cognitive behavioural therapy (CBT), family therapy or interpersonal therapy.^{[8][9]}
- Consider the need for and role of family intervention and treatment (especially for children and adolescents).^[9]
 - Family therapy is essential for recovery for children, youth and adults who are family-centred. In most cases, recovery will be limited and brief without successful family therapy. The most important aspects of family therapy are the involvement of all family members, and treatment by an experienced clinician trained in family therapy for eating disorders. Modalities include family intervention, therapeutic letter writing, individualized therapy for family members and the Maudsley method.
 - The Maudsley method involves training parents in the management of their child's condition. It is primarily used for very young patients and parents with abundant time and resilience. The Maudsley method preferably begins with psychoeducation, followed by family intervention. It concludes with family systems therapy, which trains the family to care for children who have not gone through or are going through individuation from the family (usually in a multifamily setting).
 - Families with a history of trauma (e.g., abuse, neglect or death) may require an alternate approach by the family therapist.
- Set stepwise nutritional goals in conjunction with an expert dietician.
- Recommend nutritional supplements (e.g., Ensure, Boost) to achieve weight gain if not possible with food. Trained support at mealtime will improve success.^[10] Tube feeding may be necessary if oral feeding fails.
- Suggest limiting exercise. A supervised graded exercise plan (that may include limited movement like nonexercise yoga) can reduce anxiety while not interfering with the rate of weight gain.^{[9][11]}
- Recommend warming by means of a warm room or warming vest, which may reduce anxiety, hasten recovery and improve the chance of long-term recovery.^[12]
- Monitor binge and purge behaviour and set goals for normalization, e.g., a gradual tapering of laxatives.

- Note that once body fat has been normalized, psychotherapy becomes more effective.

Pharmacologic Choices

[Table 3](#) lists drugs used in the management of anorexia nervosa.^[13]

Prokinetic Agents

Food restriction and weight loss frequently lead patients with anorexia to experience gastroparesis. The prokinetic agents **domperidone** and **metoclopramide** can reduce the feeling of fullness caused by decreased intestinal motility during the early stages of feeding.^[14] Unless the antiemetic effect of metoclopramide is needed, domperidone is the preferred agent because it has a lower incidence of extrapyramidal side effects.

On rare occasions when the feeling of fullness is limiting a patient's treatment, and domperidone or metoclopramide is not effective, consider switching to **erythromycin** (must be compounded; available only in powder form in Canada) or **azithromycin**.^{[13][15][16]} Note, however, that tachyphylaxis develops to the motility effects of these antibiotics after 1–4 weeks of use.^[13]

Prokinetic agents have the potential to cause QT_c prolongation, especially in patients taking other QT_c—prolonging drugs, or with other risk factors such as hypokalemia or hypoglycemia.^{[13][17]} An ECG might be indicated prior to initiation and after dose adjustments. If domperidone is used, the QT_c should be measured before and 1 week after the start of the therapy. If the QT_c increases by greater than 50 msec, domperidone should be discontinued.

Constipation is also a frequent problem associated with weight loss. **Prucalopride** is a newer serotonin-4 (5-HT₄) receptor agonist that can help normalize colonic function and treat constipation in patients with anorexia nervosa.^[18] Unlike previously marketed 5-HT₄ agonists, prucalopride does not act at other receptors or cause cardiac complications associated with a prolonged QT_c interval.

Other Therapies

In patients with anorexia nervosa, **zinc gluconate** can increase the rate of weight gain irrespective of the serum zinc measurement.^[19] Nausea affects about 2% of patients who take oral zinc; taking with food reduces this adverse effect.

The second-generation antipsychotic **olanzapine** has been shown in some clinical trials to decrease delusional thinking, anorexic rumination and obsessiveness in low doses (2.5–10 mg daily). This may result in improved motivation and weight gain, even in chronic anorexia nervosa.^{[20][21][22][23]} However, not all trials showed this psychological benefit.^{[20][24][25][26][27]} Completion rates and adherence were also low in some trials.^{[20][26][27]} Weight gain with olanzapine treatment was either modest or did not differ from placebo; thus, it appears that olanzapine in anorexia patients does not cause the rapid weight gain it is known to cause in people of normal weight.^{[13][24][25][26][27]} This is important because patients with anorexia nervosa would not use the medication if weight gain were rapid.

Olanzapine should not be used long term unless the benefits outweigh the risk of developing a movement disorder as an adverse effect. Duration of therapy is approximately 3–4 months and to a maximum BMI of 17 kg/m², until it is no longer needed for weight gain.^[27] Olanzapine can cause metabolic effects such as dyslipidemia. Diabetes due to reduced insulin sensitivity is also a potential adverse effect, but has not been observed in this patient group to date.

Cyproheptadine at bedtime may be useful, particularly in chronic anorexia nervosa. It may cause modest weight gain and can be used as a hypnotic.^{[28][29]}

Benzodiazepines (e.g., **clonazepam**) lack demonstrable efficacy for treatment of anxiety in anorexia nervosa and carry a risk of misuse.^[8] If severe anxiety is present, see [Anxiety Disorders](#). Take care to avoid dependence (see [Benzodiazepines \(CPhA Monograph\)](#)). The second-generation antipsychotic **quetiapine** is often used to manage anxiety in this setting. It is more effective than benzodiazepines and less likely to be associated with dependence.^{[30][31]} Use quetiapine with caution, as it may increase anxiety due to a sense of loss of control, or produce withdrawal symptoms if used long term.^{[32][33]} It may also cause extrapyramidal side effects because it is often used on a sliding scale with extra doses for marked anxiety.

SSRI antidepressants such as **fluoxetine** should be used only for coexistent depression, purge behaviour or anxiety disorders including obsessive-compulsive disorder (OCD), and if cardiac status is stable (e.g., QT_c less than 450 msec; see Bulimia Nervosa, [Antidepressants](#)).^[34]

Give **thiamine** at the beginning of feeding to prevent the development of Wernicke-Korsakoff syndrome (encephalopathy).

Ondansetron is not commonly used in controlling nausea or vomiting associated with eating disorders.^[35]
^[36]

Therapeutic Tips

- Hypoglycemia can occur when the patient starts eating because liver glycogen stores are depleted. Measure blood glucose if confusion occurs and 2 hours after meals for the first 1–2 days of feeding.^[37]
- Treat chronic laxative abuse by slowly tapering laxatives over months to years.^[38] Prucalopride is effective in normalizing bowel movements in many patients with chronic laxative abuse.
- Refeeding syndrome is a serious complication that can occur when patients are treated for eating disorders.^[39] Serious electrolyte disturbances, the hallmark of which is hypophosphatemia, can occur when patients are fed after a prolonged period of minimal caloric intake or starvation. Careful management by experienced clinicians is required.
- Patients with anorexia nervosa can become pregnant when amenorrheic.^[40]
- Refusal of treatment is common. Carefully reassess the treatment plan if this occurs.^[41] Family therapy is particularly important for the treatment of children and adolescents.

- Suicidality, worsening depression and/or an inability to gain weight are indicators for referral to an eating disorders specialist. Early treatment has a greater chance of success. Outpatient, residential or inpatient eating disorder treatment may be required.^[42]

Bulimia Nervosa

Therapeutic Choices

[Figure 2](#) illustrates the management of bulimia nervosa. Patients should also be assessed and treated for suicidal ideation and depression.

Nonpharmacologic Choices

- CBT and interpersonal therapy are helpful in addressing cognitive and emotional issues and reinforcing normal eating behaviour. Monitor the patient's progress.^[4]
- Psychoeducational groups addressing nutritional and psychological issues can enhance individual therapy.^{[43][44]}
- A self-help approach offers an accessible early intervention for both adolescent and adult patients. Book- and internet-based CBT programs, many guided by email support from a clinician, are available for the treatment of bulimia nervosa, binge eating and eating disorders not otherwise specified. Studies have shown that in patients with no coexistent psychiatric conditions, guided self-help therapy may be effective in reducing episodes of bingeing, purging and vomiting, and in improving anxiety and quality of life within 2–6 months post-treatment.^{[45][46][47][48][49][50]}

Pharmacologic Choices

[Table 4](#) lists drugs used in the management of bulimia nervosa.

Antidepressants

Antidepressants are effective in reducing binge-eating episodes by ≥50% in two-thirds of patients. **SSRIs**, **venlafaxine** or **trazodone** can be used; however, **fluoxetine** is recommended as it is supported by the most evidence.^{[8][20][51][52][53]} There is no correlation between the presence of comorbid depression and the antibulimic effects of the antidepressant, nor is there any evidence that one antidepressant is more effective than another.^[54] Trazodone can be useful in patients with insomnia associated with bulimia nervosa.^[20] Due to toxicity and poor adherence, **TCA**s and **MAOI**s should not be prescribed.^[20] **Bupropion** is contraindicated as it has been associated with an increased risk of seizure in patients with eating disorders.^[8] If therapy is effective, continue for at least 6 months, and preferably 1 year.

Therapeutic Tips

- Purging can reduce or prevent drug absorption. When pharmacotherapy is not effective, ask the patient about the timing of both purging behaviour and dose administration.
- A temporary worsening of binge and purge behaviour often occurs with significant life stress or during psychological treatment, e.g., when a long-suppressed traumatic event such as sexual abuse is uncovered and addressed. This does not indicate a worsening in the patient's overall condition.
- Treatment with more than one antidepressant at the same time is not recommended as there are no proven advantages, and potentially increased adverse effects and cost.
- Response to medication will vary from patient to patient. A trial of several agents may be needed to find one that is effective.
- When an antidepressant is effective, continue for 6–12 months before considering tapering.
- When discontinuing antidepressants, taper the dose gradually to prevent the development of antidepressant discontinuation syndrome. For more information, see Antidepressant Discontinuation Syndrome in [Depression](#).
- Treatment of any psychiatric comorbidity is necessary for recovery.
- Treatment by a multidisciplinary team may be necessary.

Binge-Eating Disorder

Therapeutic Choices

Binge-eating disorder (BED) is diagnosed when a patient has recurrent episodes of eating excessive amounts of food. These are associated with a feeling of loss of control but relief from stress during the binge and with increased anxiety and decreased self-esteem following the binge (see [Table 1](#)).

BED occurs in all genders and is more common in patients in their 30s. It is usually a chronic condition that goes unreported by the patient due to shame. In some patients, it leads to or worsens obesity. BED is a type of addiction and is more common in those with a family history of addictions.^{[55][56]} Patients with BED may report an association between stress, time of day and time of cycle (if menstruating) with the binge.^{[57][58]}

BED, distinct from bulimia nervosa, has no compensatory behaviour that follows the binge that would reduce weight gain. [Figure 3](#) illustrates the management of BED.

Nonpharmacologic Choices

- Determine the factors that make binge eating more likely (e.g., changes in mood, stressors, life situations, eating behaviour) and help the patient deal with those factors by other means.

- Recommend that the patient consume 3 meals a day with a source of protein in each meal, and that they avoid large amounts of simple sugars.
- Focus on decreasing the size or likelihood of the binge by inserting water and a snack before the usual time of the binge.
- Avoid:
 - a calorie-restricted diet
 - a diet that does not meet the patient's lifestyle and comorbid illness (e.g., allergies, diabetes, traditional dishes)
- The standard therapy for BED is CBT; however, depending on the experience of available therapists, other therapies such as dialectical behaviour therapy may be tried. Self-help resources for CBT are also available.

Pharmacologic Choices

Selective Serotonin Reuptake Inhibitors

SSRIs have demonstrated efficacy in increasing binge-eating abstinence. They also effectively reduce binge-eating frequency and eating-related obsessions and compulsions.^[59] SSRIs may be beneficial in patients with concomitant mood disorders. SSRIs have been shown to be ineffective for weight loss.^[60] Fluoxetine may be advantageous as it is generally well-tolerated with minimal adverse effects.^[60]

Amphetamine-Based Agents

Lisdexamfetamine is the only medication approved by Health Canada for the treatment of BED. Systematic reviews have shown its efficacy in reducing binge-eating frequency and binge-eating-related obsessions and compulsions.^[61] Additionally, it has been shown to reduce weight and appetite.^[61] Lisdexamfetamine is well tolerated in patients with BED and offers the flexibility of gradual titration due to its broad dosing spectrum.

Anticonvulsants

Topiramate has been suggested by some authors and in clinical guidelines as a potential treatment for binge-eating disorder.^{[8][20][62][63]} It has shown efficacy for weight loss and reducing binge-eating frequency in clinical trials and a systematic review.^{[8][20][62]} However, withdrawal rates in the trials were high.^{[8][62]} Furthermore, patients receiving topiramate experienced significantly more side effects, particularly cognitive difficulties and depression, compared to placebo.^{[8][62]} The use of topiramate is therefore not recommended.

Therapeutic Tips

- Rule out other causes of bingeing, including major depressive disorder, bulimia nervosa, Kleine-Levin syndrome, Prader-Willi syndrome, major tranquilizers, mood stabilizers (especially lithium and antidepressants), marijuana, hypoglycemia, zinc deficiency and iron deficiency.
- Diagnose and treat family dysfunction and comorbid disorders, including major depressive disorder, addiction, diabetes mellitus (especially when accompanied by episodes of hypoglycemia), post-traumatic stress disorder and attention deficit hyperactivity disorder.
- Set objectives and goals.
 - Define the factors that constitute a binge (e.g., food volume, emotional state) so the patient does not feel unnecessarily unsuccessful for eating a slightly larger amount.
 - Define what factors are as important to the patient as decreasing bingeing. This might include improving self-esteem, overcoming depression or anxiety, reducing excess body fat, and controlling related disorders like diabetes.
 - Determine what treatments the patient has tried, what they would like to try and what they are willing to try.

Avoidant/Restrictive Food Intake Disorder

Therapeutic Choices

Avoidant/restrictive food intake disorder (ARFID) is characterized by avoidance of certain foods. This may depend on the texture, taste, smell or remembered associations of the food. Although many children and adults are “picky eaters,” if the food avoidance compromises health, emotional or social well-being, it may be ARFID.^[64] For example, teenagers often seek treatment for ARFID if their food avoidance compromises their social life. [Figure 4](#) illustrates the management of ARFID.

Nonpharmacologic Choices

Assess and treat the associated health, emotional and social status of the patient. Deficiencies should be corrected if present. Iron, zinc, B₁₂, folate and vitamin C deficiencies are most important as they can alter appetite, mood, energy levels, taste and food preference.^[65] Serum zinc is not a good measure of brain zinc; therefore, an oral supplement should be tried if the diet is deficient. This is often the case with a diet that is low in red meat or dairy products.^{[65][66]}

Concurrent psychological disorders should be treated. Sources of stress, such as family dysfunction, school or work, should be ameliorated as well.

CBT is a commonly used treatment for ARFID. CBT strategies focus on decreasing food avoidance, increasing food exposure, and targeting cognitions underlying the diet restrictions, such as fear of

vomiting or choking. CBT interventions can include goal setting, psychoeducation, self-monitoring, graded food exposure and anxiety management. Studies have shown that CBT may improve various outcomes in ARFID, including weight gain, nutritional intake and variety of accepted foods.^[67]

Desensitization to the avoided food is of primary importance. To desensitize, begin with the least aversive foods. If texture is important, gradually prolong the presence of the food on the tongue or mix foods of different textures. If taste or smell is of concern, use very small amounts or mix the food with other foods or condiments.^{[66][68]} Desensitization should be done in a relaxing environment. Methods of distraction can be incorporated. For example, the patient could talk with family at mealtimes without focusing on eating, or eat with friends at a party or event.

Pharmacologic Choices

Medications should be used to treat co-occurring conditions like OCD (see [Obsessive-Compulsive Disorder](#)), anxiety disorders (see [Anxiety Disorders](#)) or major depressive disorder (see [Depression](#)). ARFID is more common in children with autism spectrum disorder.^[69]

Therapeutic Tips

- Start with the least noxious foods for desensitization.
- It is not necessary for the patient to eat every food or texture. The goal is normalization of nutrition, stress and lifestyle.
- Patients who are underweight should be encouraged to eat larger volumes of preferred foods in the early stages of treatment, before increasing dietary variety in later stages.^[65]
- Exercise restriction or limitation may be necessary initially if the patient is malnourished, has blood glucose or electrolyte abnormalities, or has cardiovascular complications.^[70]
- Nutritional supplements (e.g., Ensure, Boost) or tube feeding may be required initially if oral feeding fails to achieve weight gain; however, this should be a temporary measure, as long-term use may hinder the development of skills related to eating.^{[65][71]}
- In patients with severe malnutrition, hypoglycemia and refeeding syndrome may occur when eating/feedings resume. Monitor blood glucose and electrolytes and give thiamine if necessary.^{[65][71]}

Choices during Pregnancy and Breastfeeding

Eating Disorders and Pregnancy

Amenorrheic patients with anorexia nervosa may ovulate and, unless effective contraception is used, pregnancy can occur.^{[72][73][74][75][76]} Medications, including oral contraceptives, may be missed, vomited

or not absorbed.^[77]

Eating disorder symptoms improve during pregnancy in two-thirds of cases, but worsen in one-third.^{[78][79][80][81][82]} Patients with eating disorders have a higher reported prevalence of some pregnancy-related complications. These may include preeclampsia, miscarriage, preterm birth, cesarean delivery, small or large for gestational age babies and perinatal death.^{[83][84][85]} They also have a higher risk of anxiety and depression during pregnancy and postpartum.^{[83][84][86]} However, in most patients with eating disorders, the delivery is usually normal and the newborn is healthy.^{[85][87]}

Note that children of parents with eating disorders are often aware of the disorder and may take on a caregiver's role within the first few years of life. They also have an increased likelihood of developing disordered eating and body image themselves.^[88]

Management

- Assess for anxiety, depression or suicidality during pregnancy and postpartum.^{[89][90]}
- Reassess all medications regarding their risk in pregnancy and discontinue or change drug therapy as required. An individual risk-benefit assessment should guide decision-making.^{[91][92]} As with any medical condition during pregnancy, use nonpharmacologic measures preferentially when possible.
- Biochemistry must be measured (see [Investigations](#)).
- Follow as a high-risk pregnancy unless the eating disorder is mild.
- A dietitian should make recommendations for diet and nutrient supplements.

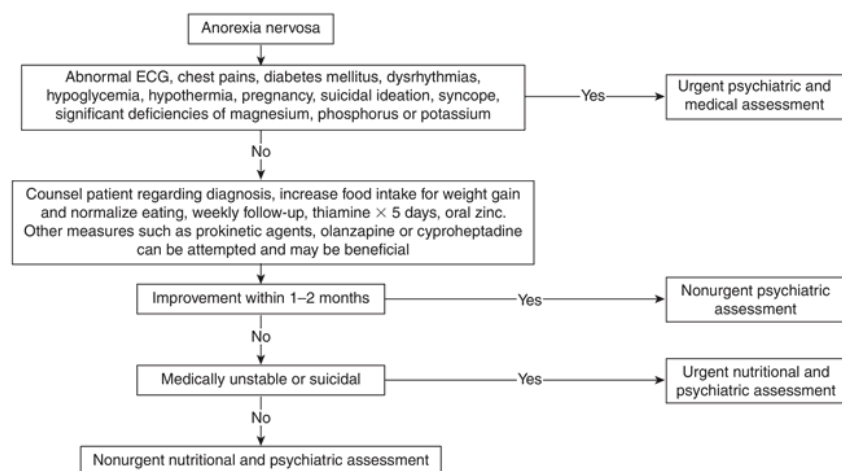
Eating Disorders and Breastfeeding

If the eating disorder (particularly anorexia nervosa) is classified as severe, breast milk production can be affected. Nevertheless, most patients with eating disorders can successfully breastfeed their babies. Refer breastfeeding patients with eating disorders to a dietitian to optimize maternal and infant nutrition.

A discussion of general principles on the use of medications in these special populations can be found in [Drugs Use during Pregnancy](#) and [Drug Use during Breastfeeding](#). Other specialized reference sources are also provided in these appendices. Specific guidance on the use of antidepressant, antipsychotic and prokinetic agents during pregnancy and breastfeeding can be found in [Depression](#), [Schizophrenia and Related Psychotic Disorders](#) and [Nausea and Vomiting](#) respectively.

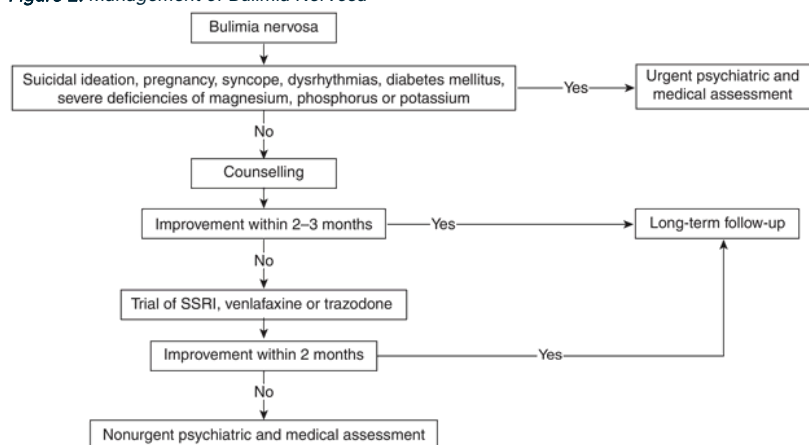
Algorithms

Figure 1: Management of Anorexia Nervosa



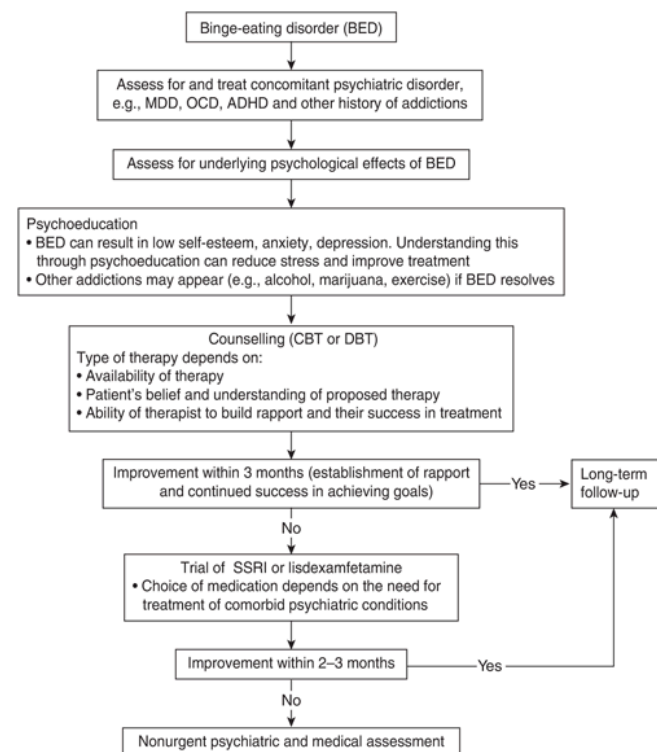
Abbreviations: ECG = electrocardiogram

Figure 2: Management of Bulimia Nervosa



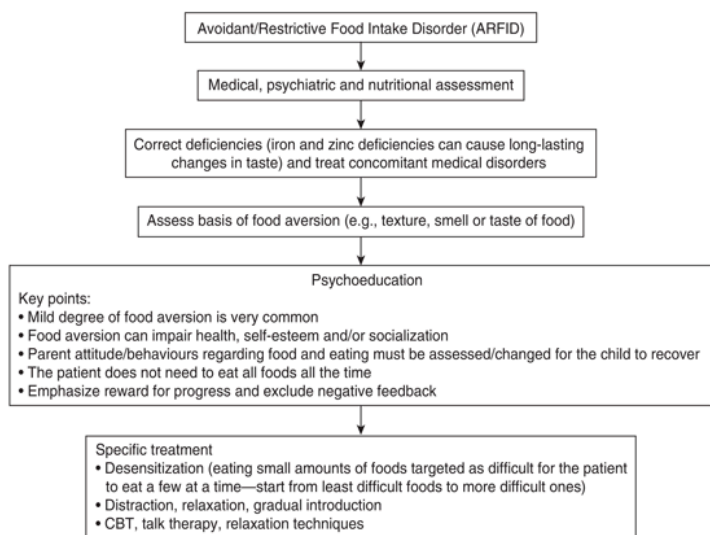
Abbreviations: SSRI = selective serotonin reuptake inhibitor

Figure 3: Management of Binge-Eating Disorder



Abbreviations: ADHD = attention deficit hyperactivity disorder; CBT = cognitive behavioural therapy; DBT = dialectical behaviour therapy; MDD = major depressive disorder; OCD = obsessive-compulsive disorder; SSRI = selective serotonin reuptake inhibitor

Figure 4: Management of Avoidant/Restrictive Food Intake Disorder



Abbreviations: CBT = cognitive behavioural therapy

Drug Tables

Table 3: Drugs Used in Anorexia Nervosa

Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Drug Interactions
Drug Class: Antihistamines				

Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Drug Interactions
cyproheptadine generics \$15–40	Hypnotic	2–4 mg QHS PO	Drowsiness, weight gain	Avoid use with alcohol or other CNS depressants. Enhances the effects of anticholinergic agents, e.g., antihistamines, TCAs , oxybutynin. Decreases the pharmacologic effects of SSRIs .

Drug Class: Antipsychotics, second-generation

olanzapine Zyprexa , Zyprexa Zydis , generics < \$15	Weight gain promotion	Start at 0.625–1.25 mg and slowly titrate to usual dose of 2.5–5 mg/day PO Maximum: 20 mg/day	Drowsiness, EPS/akathisia, dyslipidemia. Drug reaction with eosinophilia and systemic symptoms (DRESS) is a rare but very serious adverse effect. See Drug-Induced Skin Reactions for more information regarding DRESS.	Fluvoxamine may increase olanzapine serum concentrations; other SSRIs are less likely to exhibit this interaction. Inducers of CYP1A2 , such as carbamazepine, rifampin or cigarette smoking, may decrease olanzapine serum concentrations; conversely, quitting smoking may necessitate a dose reduction to avoid toxicity. Inhibitors of CYP1A2 , such as ciprofloxacin or ketoconazole, may increase effects and toxicity of olanzapine.
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Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Drug Interactions
				Caution re: other drugs with potential EPS such as metoclopramide; avoid concurrent use.
quetiapine Seroquel, generics < \$15	Anxiety	12.5–50 mg before meals and at bedtime PO Used to manage severe anxiety in this setting	Drowsiness, EPS , QT _c prolongation.	Increased hypotension with antihypertensives; additive sedation with CNS depressants, including alcohol; inhibitors of CYP3A4 , such as erythromycin, clarithromycin, fluoxetine or ketoconazole, can increase the effects and toxicity of quetiapine. Caution with other drugs that prolong the QT _c interval.

Drug Class: Serotonin 5-HT₄ Receptor Agonists

Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Drug Interactions
prucalopride  Resotran, generics \$65–100	Constipation, intestinal motility	Initial: 2 mg daily PO; decrease if possible to 1 mg daily PO after 1– 2 months	Abdominal pain, diarrhea, headache, nausea.	Monitor therapy if combined with p- glycoprotein inhibitors such as erythromycin, ketoconazole, verapamil, cyclosporine; may increase prucalopride serum concentration and adverse effects. Prucalopride-induced diarrhea may affect the absorption of oral contraceptives.

Drug Class: Prokinetic Agents				
azithromycin Zithromax, generics \$15–40	Intestinal motility (if domperidone or metoclopramide are ineffective)	250 mg daily PO	Diarrhea, abdominal discomfort, rash, prolonged QT _c interval, dysrhythmias.	Caution with other drugs that prolong the QT _c interval. May inhibit p- glycoprotein and increase adverse effects of substrates; use caution if using concurrently with a p- glycoprotein substrate (e.g., colchicine, cyclosporine, digoxin). Does not have significant CYP - related interactions.

domperidone generics < \$15	Intestinal motility	10–20 mg 30 min before meals PO May also be given at	Diarrhea, abdominal discomfort, hyperprolactinemia, drowsiness; can prolong QT _c interval.	Opioids and antimuscarinics may antagonize action of domperidone in the gut.
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Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Drug Interactions
		bedtime to deal with residual food ingested in evening Maximum: 30 mg/day		Caution with other drugs that prolong the QT _c interval.
erythromycin generics < \$15	Intestinal motility (if domperidone or metoclopramide are ineffective)	125 mg BID PO ^[b]	Diarrhea, abdominal discomfort, prolonged QT _c interval, dysrhythmias.	Inhibitor of CYP1A2 and potent inhibitor of CYP3A4 ; may increase effect and toxicity of substrates of these isoenzymes such as isotretinoin, mirtazapine, morphine, oxycodone, quetiapine, theophylline, trazodone, tricyclic antidepressants, venlafaxine. Caution with other drugs that prolong the QT _c interval.
metoclopramide  generics < \$15	Intestinal motility, nausea	5–20 mg 30 min before meals PO	Diarrhea, abdominal discomfort, hyperprolactinemia, drowsiness, restlessness; can prolong QT _c interval. Can cause EPS (rare).	Potentiates effects of other CNS depressants, including alcohol. Caution with other drugs that prolong the QT _c interval. Additive risk of EPS with antipsychotics; avoid concurrent use.
Drug Class: Supplements, mineral				

Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Drug Interactions
zinc gluconate generics < \$15	Weight gain promotion	100 mg daily PO with meals for 2 months	Nausea; can cause a copper deficiency if taken in excess for months.	Zinc may reduce the serum concentration and efficacy of tetracycline; tetracycline doses should be taken 2–3 h before zinc.
Drug Class: Supplements, vitamin				
thiamine generics < \$15	Wernicke-Korsakoff syndrome prevention	100 mg daily IM × 5 days at start of refeeding	None expected.	None known.

^[a] Cost of 30-day supply of mean dose; includes drug cost only.

^[b] Oral formulations are unavailable in Canada; however, treatment course can be compounded using erythromycin powder. Choose suitable alternative (e.g., azithromycin), if possible

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).

Abbreviations: CNS = central nervous system; CYP = cytochrome P450; EPS = extrapyramidal symptoms; SSRI = selective serotonin reuptake inhibitor; TCA = tricyclic antidepressant

Table 4: Drugs Used in Bulimia Nervosa				
Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	
Drug Class: Selective Serotonin Reuptake Inhibitors (SSRIs)				
fluoxetine ^[b] Prozac, generics \$10-30	20–60 mg daily PO; the greatest efficacy is seen at higher doses (60 mg)	Anxiety, GI discomfort (common), sexual dysfunction, increased risk of GI bleeding.	Do not use with MAOIs ; inhibits CYP2D6 —may increase effects and toxicity of many drugs including atomoxetine, dextromethorphan, mirtazapine, risperidone, TCAs ; increased risk of GI bleeding with NSAIDs , antiplatelet agents.	
Drug Class: Serotonin Modulators				
trazodone generics < \$10	100–500 mg daily PO in single or divided doses	Sedation (common), anticholinergic adverse effects less	Avoid use with MAOIs ; caution with other serotonergic drugs such as SSRIs , SNRI; inhibitors of CYP3A4 , such as erythromycin, clarithromycin or	

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions
		common than with TCAs .	ketoconazole, can increase effects and toxicity of trazodone.
Drug Class: Serotonin-Norepinephrine Reuptake Inhibitors (SNRIs)			
venlafaxine  Effexor XR , generics < \$10	75–150 mg daily PO	Nausea, drowsiness, nervousness, dizziness, dry mouth, hypertension at doses >300 mg.	Inhibitors of CYP2D6 or CYP3A4 , such as erythromycin, clarithromycin, fluphenazine or ketoconazole, can increase effects and toxicity of venlafaxine; do not use with MAOIs ; caution with trazodone (serotonin syndrome).

^[a] Cost of 30-day supply of mean dose; includes drug cost only.

^[b] For other SSRIs, see [Depression](#).

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).

Abbreviations: CYP = cytochrome P450; GI = gastrointestinal; MAOI = monoamine oxidase inhibitor; NSAID = nonsteroidal anti-inflammatory drug; TCA = tricyclic antidepressant

Table 5: *Drugs Used in Binge-Eating Disorder*

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions
Drug Class: Selective Serotonin Reuptake Inhibitors (SSRIs)			
fluoxetine ^[b] Prozac, generics \$10–\$30	20–60 mg daily PO	Anxiety, GI discomfort (common), sexual dysfunction, increased risk of GI bleeding.	Do not use with MAOIs ; inhibits CYP2D6 —may increase effects and toxicity of many drugs including atomoxetine, dextromethorphan, mirtazapine, risperidone, TCAs ; increased risk of GI bleeding with NSAIDs , antiplatelet agents.
Drug Class: Amphetamine-Based Agents			
lisdexamfetamine Vyvanse , generics \$75–\$150	30 mg daily PO; may increase by 20 mg Q 7 days Maximum: 70 mg daily PO Greatest efficacy seen with doses	Anorexia, insomnia, weight loss, irritability, dizziness, weepiness, headache, abdominal pain, “zombie-like” effects, psychotic reactions (such as hallucinations), agitation,	Avoid use with MAOIs such as phenelzine, tranylcypromine and moclobemide, as they can increase hypertensive effect; caution with other serotonergic drugs such as SSRIs and SNRI; theophylline may increase risk of tachycardia, palpitations, dizziness, weakness;

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions
	of 50–70 mg daily	tachycardia, hypertension, growth failure, rebound hyperactivity, leukopenia, blood dyscrasias.	weak inhibitor of CYP2D6 ; TCAs and antipsychotics, such as chlorpromazine, fluphenazine, can increase effects and toxicity of lisdexamfetamine.

^[a] Cost of 30-day supply of mean dose; includes drug cost only.

^[b] For other SSRIs, see [Depression](#).

Abbreviations: CYP = cytochrome P450; GI = gastrointestinal; MAOI = monoamine oxidase inhibitor; NSAID = nonsteroidal anti-inflammatory drug; TCA = tricyclic antidepressant

Suggested Readings

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[Crone C, Fochtmann LJ, Attia F et al. \(2023\). *The American Psychiatric Association practice guideline for the treatment of patients with eating disorders, 4th edition*. \[PDF file\]. Available from: <https://psychiatryonline.org/doi/epdf/10.1176/appi.books.9780890424865>.](#)

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